

Breast cancer

Every decision runs off **receptor subtype** (HR+/HER2–, HER2+, TNBC — and now **HER2-low / ultralow**) crossed with **stage and setting** · AJCC 8th anatomic + prognostic staging

Compiled from the *Stanford Oncology Handbook 2025, Breast Cancer* (Schultz C, Khan S, Sidhu K, Yang M, Dalal N, Tsai J) · NCCN · primary trial reports verified against PubMed, September 2026.

WORKUP

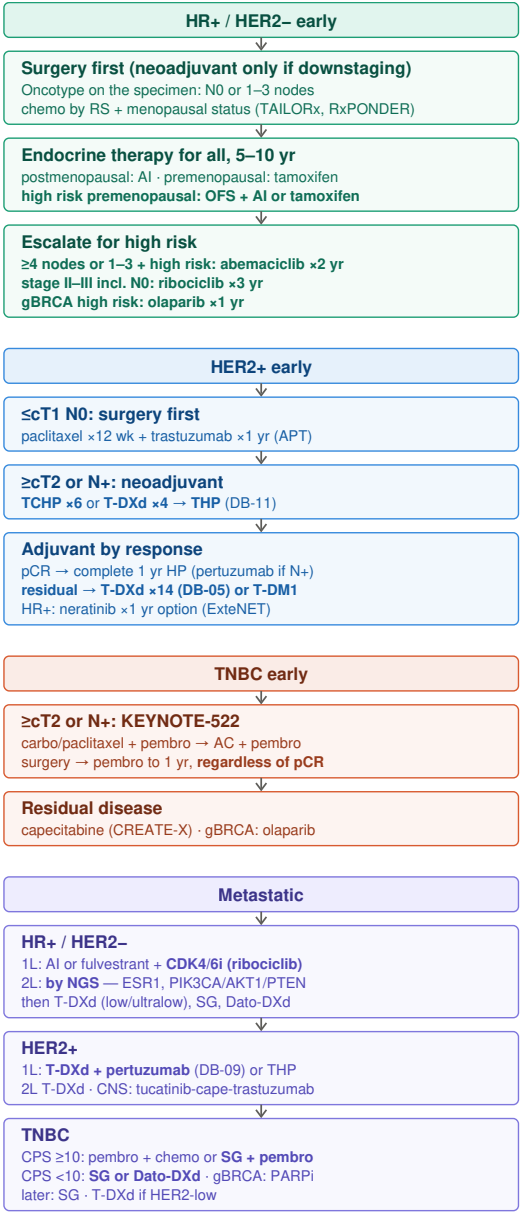
- **Bilateral diagnostic mammogram + ultrasound**; MRI or contrast-enhanced mammogram if dense tissue, young, or extent unclear
- **Core biopsy of breast (and suspicious node) with clip placement** — clip the node before neoadjuvant therapy
- **ER/PR, HER2 (IHC, reflex ISH if 2+), grade, Ki-67** on every invasive cancer
- **Stage I–IIB: no routine staging imaging** — CBC/CMP; bone scan, CT only for symptoms, raised alk phos or abnormal LFTs
- **Clinical stage III: CT C/A/P + bone scan, or PET/CT**
- **Stage IV / recurrence: biopsy the first metastasis** (receptors drift, HER2-low status can change); brain MRI for CNS symptoms · **NGS on tissue or ctDNA** for PIK3CA/AKT1/PTEN/ESR1 · germline testing if not done
- Fertility counseling before chemotherapy in premenopausal patients

STAGING — AJCC 8TH

T1 / T2 / T3	≤20 mm (T1mi ≤1, T1a ≤5, T1b ≤10) / >20–50 mm / >50 mm
T4	a chest wall · b skin edema/ulcer/satellite nodules · c both · d inflammatory
pN1 / pN2 / pN3	1–3 / 4–9 / ≥10 axillary nodes (or infraclavicular, supraclavicular). pN1mi >0.2–2 mm; pN0(i+) ≤0.2 mm
cN2 / cN3	fixed/matted axillary or IMN alone / infraclavicular, IMN + axillary, or supraclavicular
Prognostic stage	Grade, ER, PR, HER2 modify the anatomic stage — triple-negative is upstaged, HR+/HER2+ often downstaged. T1–2 N0 HR+/HER2– with Oncotype RS <11 → stage IA . Pathologic prognostic stage does not apply after neoadjuvant therapy
After neoadjuvant	yc / yp prefix; pCR = ypT0/is ypN0 — drives adjuvant escalation

DCIS: lumpectomy + RT, mastectomy, or **lumpectomy alone if small and age >70**; margin ≥2 mm; ER+ → tamoxifen ×5 yr (NSABP B-24) or low-dose 5 mg ×3 yr. LCIS: florid/pleomorphic → excise to negative margins; classic does not need them.

TREATMENT PATHWAY BY SUBTYPE



RECEPTOR DEFINITIONS — ASCO/CAP

Result	Meaning
ER / PR ≥1%	HR-positive. ER-low (1–10%) behaves like TNBC — consider TNBC-type chemo/IO
HER2 IHC 0 — null	No membrane staining. No HER2-directed therapy
HER2-ultralow <i>IHC 0 with staining</i>	Faint, incomplete membrane staining in ≤10% of cells. T-DXd-eligible in HR+ mBC after ET (DESTINY-Breast06)
HER2-low	IHC 1+ (faint, incomplete in >10%) or IHC 2+ / ISH–. Not HER2-positive — no trastuzumab benefit — but T-DXd works (DESTINY-Breast04/06)
HER2-positive	IHC 3+ (complete, intense, >10%) or IHC 2+ with ISH amplification

Pathology reports must now distinguish 0-null from ultralow — ask for a re-read of an archival “IHC 0” before ruling out T-DXd. Scoring 0 vs 1+ is poorly reproducible; retest a metastatic biopsy.

GERMLINE TESTING

- **ASCO/SSO 2024: offer BRCA1/2 testing to every patient diagnosed at ≤65** (stage I–III or de novo IV). Over 65: if TNBC, male, PARP-inhibitor candidate, or suggestive personal/family history. NCCN uses ≤50 plus family/personal criteria
- **Testing is treatment-changing:** adjuvant olaparib (OlympiA), PARP inhibitor in mBC, and surgery choice (bilateral mastectomy, RRSO)
- **High penetrance:** **BRCA1** (TNBC-enriched) · **BRCA2** (HR+, male BC) · **PALB2** (manage like BRCA) · **TP53** (avoid RT if possible) · PTEN · CDH1 (lobular + diffuse gastric)
- **Moderate penetrance:** **CHEK2, ATM** — annual MRI screening; **risk-reducing mastectomy is not indicated on the gene alone**; RT is safe
- Offer a multigene panel — not BRCA alone — when history suggests other genes

LOCAL THERAPY — DE-ESCALATION

Question	Evidence
1–2 positive SLN, BCS + RT	Skip ALND (Z0011: 10-yr OS 86.3 vs 83.6%). AMAROS: axillary RT = ALND for control (10-yr axillary recurrence 0.93 vs 1.82%), half the lymphedema
cN0, negative axillary US	SLNB can be omitted: SOUND (≤2 cm, 5-yr DDFS 98.0 vs 97.7%) · INSEMA (≤5 cm, BCS; 5-yr iDFS 91.9 vs 91.7%) — axillary recurrence slightly higher
RT after BCS, age ≥65, low risk	PRIME II (HR+, N0, ≤3 cm, on ET): omitting RT raised 10-yr local recurrence 9.5% vs 0.9%, OS identical (80.8 vs 80.7%). EUROPA (≥70, luminal A): RT alone had better QoL than ET alone at interim — recurrence data pending
ALND still indicated	≥3 positive nodes, matted nodes, inflammatory/T4, node-positive after neoadjuvant

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EARLY HR+ / HER2–

Question	Evidence
Chemo? N0	TAILORx : RS 11–25 → ET alone non-inferior (9-yr iDFS 83.3 vs 84.3%). Age ≤50 with RS 16–25: chemo helps. RS ≥26 → chemo
Chemo? 1–3 nodes	RxPONDER (RS ≤25): postmenopausal — no benefit ; premenopausal — 5-yr iDFS 93.9 vs 89.0% (likely an OFS effect)
OFS	SOFT/TEXT : OFS + exemestane best DFS. Final 15-yr: OS gain confined to high risk — age <35: 82.5 vs 77.9 vs 68.1% (E+OFS / T+OFS / T)
Abemaciclib ×2 yr <i>monarchE</i>	≥4 nodes, or 1–3 + grade 3, ≥5 cm or Ki-67 ≥20%. 7-yr iDFS 77.4 vs 70.9%; OS now significant (HR 0.84; 86.8 vs 85.0%)
Ribociclib ×3 yr <i>NATALEE</i>	Stage IIB–III, or IIA N0 with grade 3 (or grade 2 + high genomic risk/Ki-67 ≥20%). 5-yr iDFS HR 0.72 (+4.5 points); OS not yet significant
Olaparib ×1 yr <i>OlympiA</i>	gBRCA, high-risk HER2–. 6-yr OS 87.5 vs 83.2% (HR 0.72)

EARLY HER2+

Setting	Evidence
Small, N0	APT (single-arm, ≤3 cm): TH → 10-yr iDFS 91.3%. ATEMPT : T-DM1 ×1 yr — 3-yr iDFS 97.8% but no less toxicity (46 vs 47%)
Neoadjuvant	NeoSphere: pertuzumab raised pCR 29 → 46%. DESTINY-Breast11 : T-DXd ×4 → THP vs ddAC → THP, pCR 67.3 vs 56.3% , grade ≥3 AE 38 vs 56%; T-DXd alone was inferior and closed
Residual disease	KATHERINE : T-DM1 ×14, iDFS HR 0.50, 7-yr OS 89.1 vs 84.4%. DESTINY-Breast05 (high risk): T-DXd vs T-DM1 , 3-yr iDFS 92.4 vs 83.7% (HR 0.47); ILD 9.6%
Pertuzumab adjuvant	APHINITY : iDFS HR 0.81; benefit is node-positive only (8-yr 86.1 vs 81.2%); OS not significant
Extended	ExteNET : neratinib ×1 yr, 5-yr iDFS 90.2 vs 87.7%; gain in HR+; no OS benefit ; 40% grade 3 diarrhea without prophylaxis

EARLY TNBC

Trial	Evidence
KEYNOTE-522	Stage II–III: pCR 64.8 vs 51.2% ; 5-yr OS 86.6 vs 81.7% . Give pembrolizumab regardless of PD-L1 or pCR
CREATE-X	Residual HER2– disease: capecitabine; TNBC OS HR 0.52
OlympiA	gBRCA with residual TNBC or ≥pT2/pN1: olaparib ×1 yr

METASTATIC HR+ — 1L CDK4/6 INHIBITORS

Trial	PFS (mo)	OS (mo)
MONALEESA-2 ribo + letrozole	25.3 vs 16.0	63.9 vs 51.4 , HR 0.76 — sig
MONALEESA-3 ribo + fulvestrant	20.5 vs 12.8	53.7 vs 41.5 , HR 0.73 — sig
MONALEESA-7 ribo, premenopausal	23.8 vs 13.0	42-mo OS 70.2 vs 46.0%, HR 0.71 — sig
MONARCH 2 abema + fulvestrant	16.4 vs 9.3	46.7 vs 37.3 , HR 0.76 — sig
MONARCH 3 abema + NSAI	28.2 vs 14.8	66.8 vs 53.7, HR 0.80 — NS
PALOMA-2 / -3 palbo + letrozole / fulv	24.8 vs 14.5 / 9.5 vs 4.6	NS in both

Ribociclib is the only CDK4/6i with OS benefit in every phase 3 — the preferred 1L agent. HR+/**HER2+** **mBC**: PATINA added palbociclib to maintenance HP + ET after induction — PFS 44.3 vs 29.1 mo (HR 0.75), OS pending.

METASTATIC HER2+

Line	Evidence
1L	DESTINY-Breast09 : T-DXd + pertuzumab vs THP, PFS 40.7 vs 26.9 (HR 0.56), approved Dec 2025; OS immature, ILD 12%. CLEOPATRA (THP): OS 57.1 vs 40.8 — still reasonable
2L	DESTINY-Breast03 : T-DXd vs T-DM1, PFS 28.8 vs 6.8 , OS HR 0.64
Brain mets	HER2CLIMB : tucatinib + trastuzumab + capecitabine, OS 21.9 vs 17.4, CNS-PFS 9.9 vs 4.2 , active mets allowed · T-DXd active (DB-12, single-arm: 12-mo PFS 61.6%)

METASTATIC TNBC

Setting	Evidence
1L CPS ≥10 <i>22C3</i>	Pembro + chemo (KEYNOTE-355): OS 23.0 vs 16.1 , HR 0.73; no OS gain at CPS 1–9 · SG + pembro (ASCENT-04): PFS 11.2 vs 7.8, OS immature (approved Jun 2026)
1L, not IO candidate	Dato-DXd (TROPION-Breast02): PFS 10.8 vs 5.6, OS 23.7 vs 18.7 (May 2026) · SG (ASCENT-03): PFS 9.7 vs 6.9, OS immature
gBRCA	Olaparib (OlympiAD, PFS 7.0 vs 4.2) or talazoparib (EMBRACA, 8.6 vs 5.6) — neither improved OS
2L+	Sacituzumab (ASCENT): OS 12.1 vs 6.7 · T-DXd if HER2-low (DB-04)

METASTATIC HR+ — AFTER CDK4/6 (BIOMARKER-DRIVEN)

Target	Agent → evidence
ESR1 mut <i>acquired on AI; test ctDNA</i>	Elacestrant (EMERALD): ESR1m PFS 3.8 vs 1.9 (HR 0.55) · imlunestrant (EMBER-3): ESR1m 5.5 vs 3.8 (HR 0.62); + abemaciclib 9.4 vs 5.5 vs imlunestrant alone (combo approved Sep 2026: ESR1m, after ≥1 line ET) · vepdegestrant (VERITAC-2, ESR1m, May 2026)
ESR1 on 1L ctDNA	SERENA-6 : switch AI → camizestrant , keep CDK4/6i, when ESR1m emerges before progression: PFS 16.0 vs 9.2 (HR 0.44). Accelerated approval Sep 2026 despite a split ODAC; OS immature
PIK3CA mut	Inavolisib + palbo + fulvestrant (INAVO120; relapse on/≤12 mo of adjuvant ET): PFS 15.0 vs 7.3, OS 34.0 vs 27.0 (HR 0.67) · alpelisib + fulv (SOLAR-1): 11.0 vs 5.7, OS NS, grade 3–4 hyperglycemia 37%; almost no prior CDK4/6i
PIK3CA / AKT1 / PTEN	Capivasertib + fulv (CAPitello-291): altered PFS 7.3 vs 3.1 (HR 0.50)
PIK3CA wild-type	Gedatolisib + fulv ± palbo (VIKTORIA-1) approved Jul 2026
No target	Switch CDK4/6i + ET (MAINTAIN, postMONARCH: PFS 6.0 vs 5.3) · everolimus + exemestane (BOLERO-2, OS NS)
Endocrine-refractory	T-DXd if HER2-low/ultralow: DB-06 (chemo-naïve) PFS 13.2 vs 8.1; DB-04 (after chemo) OS 23.9 vs 17.5 · Sacituzumab (TROPICS-02): OS 14.4 vs 11.2 · Dato-DXd (TROPION-Breast01): PFS HR 0.63 · gBRCA: PARPi

HER2-LOW / ULTRALOW — T-DXD

Trial	Result
DESTINY-Breast04 <i>after 1–2 chemo lines</i>	HER2-low (IHC 1+ or 2+/ISH–), HR+ or HR–. HR+ cohort PFS 10.1 vs 5.4 , OS 23.9 vs 17.5 (HR 0.64); all patients OS 23.4 vs 16.8.
DESTINY-Breast06 <i>after ≥1 ET, chemo-naïve</i>	HR+ only. HER2-low PFS 13.2 vs 8.1 (HR 0.62); ultralow an exploratory subgroup. OS immature. Label covers low + ultralow (Jan 2025)
ILD	~11–12% in both, with deaths

METASTATIC HR+ — SEQUENCING RULES

- **Endocrine-sensitive** (de novo, or relapse >12 mo after adjuvant ET) → **AI + CDK4/6i**. **Endocrine-resistant** (on or ≤12 mo after adjuvant ET) → **fulvestrant + CDK4/6i**, or the **inavolisib triplet** if **PIK3CA-mutant**
- **Visceral crisis** → chemotherapy first; otherwise endocrine-based therapy precedes chemo. **Premenopausal/men: add GnRH agonist**
- **Re-test ctDNA at each progression** — ESR1 is acquired under AI pressure, usually absent at diagnosis

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EARLY-STAGE DOSING

Regimen	Dosing
ddAC → T	Doxorubicin 60 + cyclophosphamide 600 q14d ×4 (G-CSF) → paclitaxel 175 q14d ×4 or 80 weekly ×12
TC	Docetaxel 75 + cyclophosphamide 600 q21d ×4
KEYNOTE-522	Carboplatin AUC 5 d1 + paclitaxel 80 d1, 8, 15 + pembrolizumab 200 mg q21d ×4 → AC (60/600) + pembro q21d ×4 → surgery → pembro ×9 (1 yr total)
TCHP / TCH	Docetaxel 75 + carboplatin AUC 6 + trastuzumab 8 → 6 mg/kg ± pertuzumab 840 → 420 mg q21d ×6; HER2 therapy to 1 yr
TH <i>APT</i>	Paclitaxel 80 + trastuzumab 4 → 2 mg/kg weekly ×12, then trastuzumab 6 mg/kg q3wk to 1 yr
AC → TH(P)	AC q14d ×4 → paclitaxel 80 weekly ×12 + trastuzumab ± pertuzumab. Never HER2 agents concurrent with anthracycline
T-DM1 / T-DXd <i>residual HER2+</i>	T-DM1 3.6 mg/kg q21d ×14 · T-DXd 5.4 mg/kg q21d ×14 (DB-05)
Capecitabine	1000–1250 mg/m² BID d1–14 q21d ×6–8
Abemaciclib <i>adjuvant</i>	150 mg BID ×2 yr + ET
Ribociclib <i>adjuvant</i>	400 mg daily d1–21 q28d ×3 yr + NSAI
Olaparib	300 mg BID ×1 yr (tablets)
Endocrine	Tamoxifen 20 mg · anastrozole 1 · letrozole 2.5 · exemestane 25 mg daily · goserelin 3.6 mg SQ q4wk / leuprolide 3.75 mg q4wk

Doses mg/m² unless stated. Neratinib: loperamide prophylaxis or dose escalation from the start.

METASTATIC DOSING

Regimen	Dosing
CDK4/6 inhibitors	Ribociclib 600 mg d1–21 q28d · palbociclib 125 mg d1–21 q28d · abemaciclib 150 mg BID continuous (200 BID alone) — with AI or fulvestrant
Fulvestrant	500 mg IM d1, 15, 29, then q28d. Premenopausal: add OFS to AI or fulvestrant
Oral SERDs	Elacestrant 345 mg daily · imlunestrant 400 mg daily · camizestrant 75 mg daily + CDK4/6i
Capivasertib	400 mg BID, 4 days on / 3 off + fulvestrant
PI3Kα	Inavolisib 9 mg daily + palbociclib + fulvestrant · alpelisib 300 mg daily + fulvestrant
Everolimus + exemestane	10 mg + 25 mg daily; dexamethasone mouthwash for stomatitis
THP	Docetaxel 75 (or weekly paclitaxel 80) + trastuzumab 8 → 6 mg/kg + pertuzumab 840 → 420 mg q21d; stop taxane after ~6 cycles, continue HP
T-DXd	5.4 mg/kg q21d (HER2+, low, ultralow) ± pertuzumab 1L. Highly emetogenic — 3-drug prophylaxis
Tucatinib-cape-H	Tucatinib 300 mg BID + capecitabine 1000 mg/m² BID d1–14 + trastuzumab 8 → 6 mg/kg q21d
Sacituzumab	10 mg/kg d1, d8 q21d (± pembro 200 q3wk)
Dato-DXd	6 mg/kg q21d
PARP inhibitors	Olaparib 300 mg BID · talazoparib 1 mg daily
Bone-directed	Zoledronic acid 4 mg IV q4wk (q12wk after 1 yr) or denosumab 120 mg SQ q4wk ; dental review, Ca/vit D

SURGICAL, RADIATION AND SPECIAL SITUATIONS

- **Inflammatory (T4d): trimodality always** — neoadjuvant systemic therapy → modified radical mastectomy + ALND → chest wall + nodal RT, **even after pCR**; no BCS, delayed reconstruction
- **BCS relative contraindications:** Li-Fraumeni, active scleroderma/ lupus, prior chest RT, large tumor-to-breast ratio. **Mastectomy** for gene carriers, prior mantle RT, or preference
- **Pregnancy:** surgery any trimester; anthracycline chemo from the 2nd trimester; **no endocrine, HER2, IO or RT until after delivery**
- **Male breast cancer:** mostly HR+, BRCA2-enriched; **tamoxifen preferred**; an AI needs a GnRH agonist
- **Lymphedema** ~5% after SLNB, highest after ALND + nodal RT · **BIA-ALCL** (textured implants): cytology, then remove capsule + implant

TOXICITY AND MONITORING

- **T-DXd ILD** (10–15% across trials, fatal cases): baseline + **serial chest CT** on therapy, ask about cough/dyspnea at every visit. **Grade 1:** **hold** until resolved (same dose if ≤28 days), consider steroids. **Grade ≥2: stop permanently** + prompt systemic steroids
- **HER2 agents — cardiac:** LVEF at baseline and **q3 months**. Hold trastuzumab for ≥**16% absolute drop**, or below normal with ≥10% drop; recheck in 4 wk
- **Ribociclib: QTc** — ECG baseline (start only if QTcF <450) and ~day 14; electrolytes; **never with tamoxifen**; LFTs + CBC q2wk ×2 cycles
- **Abemaciclib: diarrhea** — loperamide at the first loose stool; CBC/LFTs q2wk ×2 mo; VTE, ILD. **Palbociclib:** neutropenia, CBC d1 + d15 of cycles 1–2
- **Alpelisib / capivasertib / inavolisib: hyperglycemia** — fasting glucose + HbA1c first, optimize diabetes; glucose weekly-to-biweekly in month 1; metformin first-line; rash (alpelisib: antihistamine prophylaxis)
- **Sacituzumab:** neutropenia (grade ≥3 ~50%), diarrhea; UGT1A1*28 homozygotes at higher risk. **Dato-DXd:** stomatitis and keratitis — steroid mouthwash + eye care
- **Tamoxifen:** VTE, endometrial cancer (annual gyn review); avoid paroxetine/fluoxetine (CYP2D6). **AI:** DEXA baseline + q1–2 yr
- **Surveillance:** H&P q6–12 mo ×5 yr, annual mammogram; no routine markers or imaging

KEY TRIALS

KEYNOTE-522	Pembro + neoadjuvant chemo, stage II–III TNBC — 5-yr OS 86.6 vs 81.7%
monarchE / NATALEE	Adjuvant abemaciclib (OS HR 0.84) / ribociclib (iDFS HR 0.72)
DESTINY-Breast04 / 06	T-DXd in HER2-low after chemo (OS 23.4 vs 16.8) / low + ultralow after ET
DESTINY-Breast03 / 09	T-DXd 2L (PFS 28.8 vs 6.8) / 1L + pertuzumab (40.7 vs 26.9)
DESTINY-Breast05 / 11	T-DXd for residual HER2+ disease / neoadjuvant T-DXd → THP
MONALEESA-2	Ribociclib 1L OS 63.9 vs 51.4 mo
INAVO120	Inavolisib triplet, PIK3CA-mut — OS 34.0 vs 27.0 mo
SERENA-6	ctDNA-guided camizestrant switch — PFS 16.0 vs 9.2 mo
ASCENT / -04	SG in refractory TNBC / + pembro 1L CPS ≥10